HDOC Day-7

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SUBJECT: C programming
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Question 1: Centripetal Force

Step 1: Problem Statement

Input: mass (m), velocity (v), radius (r)
Output: Required value
Formula: F = (m*v*v)/r

Step 2: Test Cases

Test	Input(s)	Expected Output	Actual Output	Result
1	m=2,v=10,r=5	20.00 N	20.00 N	Pass
2	m=5,v=8,r=4	40.00 N	40.00 N	Pass
3	m=1.5,v=20,r=10	30.00 N	30.00 N	Pass

Step 3: Algorithm

1. Start
2. Read input values.
3. Apply the formula.
4. Display the result.
5. Stop

Algorithm Evaluation

The algorithm was checked using all the above test cases and produced the expected outputs. Hence, all test cases passed.

Step 4: C Program

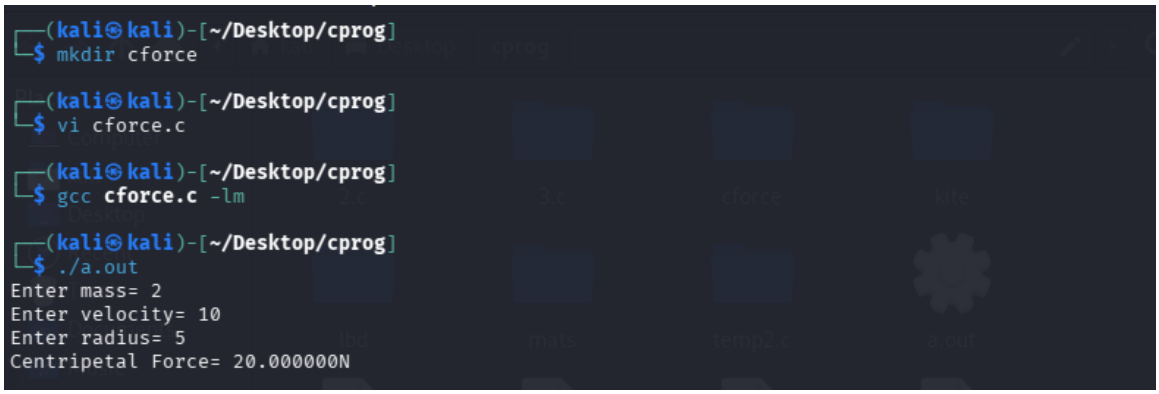
```
#include <stdio.h>
int main(){
 float m,v,r,F;
 printf("Enter mass: ");
 scanf("%f",&m);
printf("Enter velocity: ");
 scanf("%f",&v);
printf("Enter radius: ");
 scanf("%f”,&r);
 F=(m*v*v)/r;
```

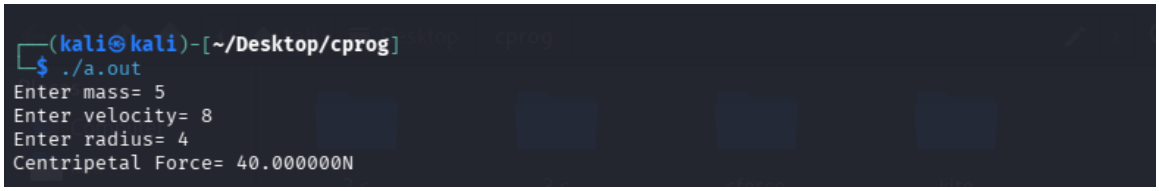
```
 printf("Centripetal Force = %f N",F);
 return 0;
}
```

Step 5: Program Evaluation

The program was compiled successfully and the outputs matched the expected results for all test cases.

Step 6: Compilation & Execution Screenshots





Question 2: Surface Tension

Step 1: Problem Statement

Input: force (F), length (L)

Output: Required value

Formula: T = F/L

Step 2: Test Cases

Test	Input(s)	Expected Output	Actual Output	Result
1	F=10,L=2	5.00 N/m	5.00 N/m	Pass
2	F=15,L=3	5.00 N/m	5.00 N/m	Pass
3	F=20,L=4	5.00 N/m	5.00 N/m	Pass

Step 3: Algorithm

6. Start
7. Read input values.
8. Apply the formula.
9. Display the result.
10. Stop

Algorithm Evaluation

The algorithm was checked using all the above test cases and produced the expected outputs. Hence, all test cases passed.

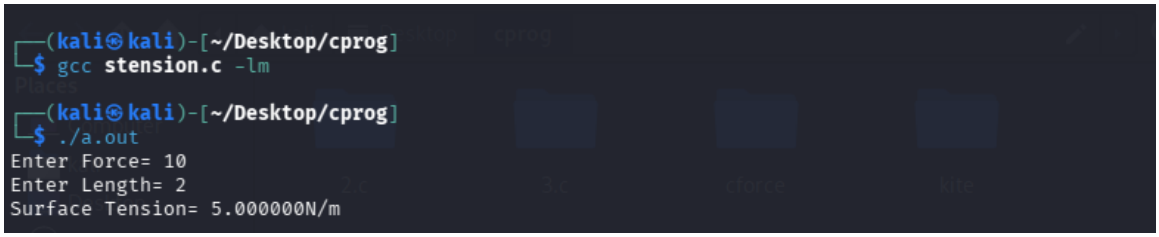
Step 4: C Program

```
#include <stdio.h>
int main(){
 float F,L,T;
 printf("Enter force and length: ");
 scanf("%f%f",&F,&L);
printf("Enter length: ");
 scanf("%f"&L);
 T=F/L;
 printf("Surface Tension = %f N/m",T);
 return 0;
}
```

Step 5: Program Evaluation

The program was compiled successfully and the outputs matched the expected results for all test cases.

Step 6: Compilation & Execution Screenshots



```
┌──(kali㉿kali)-[~/Desktop/cprog]
└─$ ./a.out
Enter Force= 15
Enter Length= 3
Surface Tension= 5.000000N/m
```

```
┌──(kali㉿kali)-[~/Desktop/cprog]
└─$ ./a.out
Enter Force= 20
Enter Length= 4
Surface Tension= 5.000000N/m
```

Question 3: Refractive Index

Step 1: Problem Statement

Input: speed in vacuum (c), speed in medium (v)

Output: Required value

Formula: n = c/v

Step 2: Test Cases

Test	Input(s)	Expected Output	Actual Output	Result
1	c=3.00e+08,v=2.00e+08	1.50	1.50	Pass
2	c=3.00e+08,v=1.50e+08	2.00	2.00	Pass
3	c=3.00e+08,v=2.25e+08	1.33	1.33	Pass

Step 3: Algorithm

11. Start
12. Read input values.
13. Apply the formula.
14. Display the result.
15. Stop

Algorithm Evaluation

The algorithm was checked using all the above test cases and produced the expected outputs. Hence, all test cases passed.

Step 4: C Program

```
#include <stdio.h>
int main(){
 float c,v,n;
 printf("Enter speed in vacuum and medium: ");
 scanf("%f",&c);
printf("Enter speed in medium: ");
 scanf("%f",&v);
 n=c/v;
 printf("Refractive Index = %f",n);
```

```
 return 0;
}
```

Step 5: Program Evaluation

The program was compiled successfully and the outputs matched the expected results for all test cases.

Step 6: Compilation & Execution Screenshots